

SAMAY SHANKAR DUBEY

samaydubey232@gmail.com | +91-8318003764 | [GitHub](#) | [LinkedIn](#)

Education

Indian Institute of Information Technology, Nagpur
B.Tech in Computer Science
Kendriya Vidyalaya, Gomti Nagar, Lucknow
Class XII: 89.8% | Class X: 96.4%

2023 – Present
CGPA: 7.2 / 10
2020 – 2023

Skills

Languages: C++, Python, JavaScript (ES6+), TypeScript, SQL
Frontend: React.js, Next.js, Tailwind CSS, Framer Motion, GSAP, D3.js, Redux Toolkit
Backend: Node.js, Express.js, REST APIs, JWT, Socket.IO, Python (subprocess)
Databases: MongoDB (Mongoose), PostgreSQL, MySQL
DevOps & Tools: AWS, Vercel, Git, GitHub, Postman, Vite, Linux

Projects

FIT QUEST – Gamified Full-Stack Fitness Platform

[GitHub](#)

Tech Stack: MongoDB, Express.js, React 18, Node.js, Socket.IO, Stripe, MediaPipe, Cloudinary, Redux Toolkit

- Designed and built a full-stack MERN fitness platform with **10 Mongoose models** (User, Workout, WorkoutLog, MealLog, Challenge, League, and more) and **200+ REST API endpoints** secured with **Helmet**, **CORS**, and **express-rate-limit** middleware.
- Implemented **JWT authentication** with HTTP-only refresh tokens, **bcrypt** password hashing, and a **Nodemailer** password-reset flow; integrated **Cloudinary** and **Multer** for scalable media uploads.
- Built a **Socket.IO** real-time layer across 4 namespaces with JWT-authenticated middleware, enabling live leaderboard pushes at **sub-80ms latency** and synced community feeds.
- Integrated a **MediaPipe** pose-analysis engine for real-time AI-powered workout form feedback.

GRAPHY – Interactive Graph Algorithm Visualizer

[GitHub](#)

Tech Stack: React 18, Node.js, Express.js, D3.js, HTML Canvas, GSAP, Framer Motion

- Implemented **9 graph algorithms** (BFS, DFS, Dijkstra, Bellman-Ford, Floyd-Warshall, Prim, Kruskal, Cycle Detection, Connected Components) using a **custom Min-Heap Priority Queue**, achieving $O((V+E) \log V)$ complexity with 120+ recorded execution traces.
- Architected a high-performance **Canvas renderer** with a decoupled **requestAnimationFrame** loop and **D3 force simulation**, delivering **sub-16ms frame updates** synchronized with GSAP and Framer Motion animations.

CryptoSentinel – Crypto Intelligence & Backtesting Platform

[GitHub](#)

Tech Stack: React 19, Node.js, Express.js, PostgreSQL, Python, FinBERT, Recharts, node-cron

- Designed a **PostgreSQL** schema with **composite B-tree indexes** on (symbol, time) and (time, headline), achieving **sub-200ms** query response times across **1,440+ Binance OHLCV** records.
- Integrated a **FinBERT** (Hugging Face Transformers) NLP pipeline via a **Python subprocess** for financial sentiment scoring; fused sentiment signals with **PostgreSQL window-function-computed MA50** to power a rule-based trading strategy.
- Automated an **ETL pipeline** via **node-cron** to ingest data from **Binance** and **Reddit** APIs daily, processing **125+ articles** per run into a centralized **sentiment_logs** table.
- Built a **backtesting engine** that joins price and sentiment data to simulate trades, reporting ROI, win rate, max drawdown, and final equity visualized with **Recharts** interactive charts.

Achievements & Coursework

- Codeforces:** [Specialist \(Rating: 1550\)](#) – solved **550+** problems; **LeetCode:** [Knight \(Rating: 1858\)](#) – solved **240+** problems.
- ICPC Online Prelims:** Secured **Rank 738** (nationwide) as part of a **3-member team** in the ICPC Online Qualifier round.
- Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Software Engineering.